Adaptive Interest-Weighted Diversification for Session-Based Recommendation under Dynamic User Intent

Reference document for the final report

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Abstract

This document gathers everything the final report needs: the research question, the simulator, the methods with their equations, the evaluation protocol, all measured results with confidence intervals, the exploratory follow-ups, negative results, limitations, reproducibility details, a claim-to-evidence map and viva notes. It is written as reference material rather than polished prose. Tags mark the status of each item, and all results apply to this simulation only.

Status tags. IMPLEMENTED + TESTED code exists and the test suite covers it; CONFIRMATORY, EVALUATION SEEDS the single frozen run on seeds 100–119; EXPLORATORY follow-ups on seeds 200–219, decided after seeing confirmatory results; TUNING SEEDS ONLY design and tuning evidence on seeds 0–4; PENDING not done yet.

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1 Research question and contribution

Central question. Can adaptive estimation of a user's changing category interests, combined with interest-weighted DPP selection, improve the balance of relevance, diversity and responsiveness to changing intent, compared with non-adaptive and non-DPP baselines?

- **RQ1 (adaptation).** Does an adaptive estimator track the true interest z_t better after drift, and at what cost when interests are stable or clicks are noisy?
- **RQ2 (diversification).** How much does DPP selection gain in diversity, and lose in relevance, for focused and for spread interests?
- **RQ3 (interaction).** Do adaptation and DPP diversification complement or substitute for each other?

Approach. A controlled, open-loop simulation in which the true latent interest z_t is known. A 2×2 factorial design crosses the profile (static cumulative mean vs adaptive bias-corrected EWMA) with the selector (Top-K vs greedy DPP), across five drift scenarios and two interest-concentration levels. Popularity and oracle-z methods serve as a floor and a diagnostic ceiling.

What the contribution is, and is not.

- **No novelty claim for the method.** Personalised quality inside a DPP kernel (Ibrahim et al., 2025), DPP kernels adapted over interactions (Liu et al., 2019), personalised diversification (Teo et al., 2016) and exponential forgetting of history (Ding and Li, 2005; Koren, 2009) already exist.
- **The contribution is a controlled empirical analysis.** It measures estimation error and adaptation lag against ground truth, and it separates the adaptation and selection effects and their interaction across drift types and concentrations. In the sources we reviewed (30, a targeted search rather than a systematic review), we did not find such a study. "Not found" does not mean "does not exist".
- **External validity is limited.** The results are synthetic and do not show real-world effectiveness (Stavinova et al., 2022).

2 Related work: what each source supports

Verification levels are recorded in `literature/literature_matrix.md`. Most entries are checked at the metadata and abstract level only. Cite them only for claims that level supports.

DPPs
$P(Y) \propto \det L_Y$; the L-ensemble $L = \mathrm{Diag}(q)\, S\, \mathrm{Diag}(q)$ splits into quality and diversity (Kulesza and Taskar, 2012). Fixed-size k-DPPs exist (Kulesza and Taskar, 2011). MAP is NP-hard (Ko et al., 1995) and hard to approximate (Kulesza & Taskar, Thm 2.9, read in full). Better-than-greedy MAP methods exist (Gillenwater et al., 2012).

DPP recommendation
Fast greedy MAP with incremental Cholesky updates and a relevance–diversity trade-off (Chen et al., 2018), our algorithm. Industrial DPP re-ranking (Wilhelm et al., 2018). DPP likelihoods for sequential models (Liu et al., 2022). RL-adapted kernels (Liu et al., 2019). A personalised DPP whose online deployment reduced click-through rate (Ibrahim et al., 2025): a warning that diversity can cost relevance.

Diversity and intent
MMR (Carbonell and Goldstein, 1998); topic diversification and intra-list similarity, where accuracy fell but satisfaction rose (Ziegler et al., 2005); intent-aware search diversification

(Agrawal et al., 2009); novelty and diversity evaluation (Clarke et al., 2008); intent-oriented recommendation diversity (Vargas et al., 2011); surveys (Kaminskas and Bridge, 2016; Wu et al., 2024).

Calibration
Matching list composition to a user's genre distribution (Steck, 2018), read in full on 3 November 2026 and used as a baseline (§9). Calibrating against recent windows instead of the full history (Lin et al., 2024), conceptually the closest work: static vs time-windowed profiles, but with calibration rather than a DPP, and on real data without ground truth.

Session-based
GRU4Rec (Hidasi et al., 2016); simple heuristics are often competitive (Ludewig and Jannach, 2018), which supports a transparent base scorer; survey (Wang et al., 2021); multi-interest models with an accuracy–diversity factor (Cen et al., 2020).

Drift
Vocabulary for sudden, gradual and recurring drift, and forgetting by fading factors and windows (Gama et al., 2014); adaptive windows (Bifet and Gavaldà, 2007).

Simulation
Latent-state user simulators as a research tool (Ie et al., 2019); the gap between simulated and real behaviour (Stavinova et al., 2022).

3 Simulator

IMPLEMENTED + TESTED Specification: `SIMULATION_DESIGN.md`. Code: `adaptdiv/catalogue.py`, `scenarios.py`, `simulator.py`, `seeding.py`, `config.py`.

3.1 Catalogue

$C = 10$ categories with $n_c = 50$ items each ($N = 500$). Centroids $\mu_c = \text{normalise}(h_c)$ with $h_c \sim \mathcal{N}(0, I_d)$, $d = 32$. Item embeddings $\phi_i = \text{normalise}(\mu_{c_i} + \sigma\varepsilon_i)$, $\varepsilon_i \sim \mathcal{N}(0, I_d/d)$, $\sigma = 1$. The mean cosine similarity is 0.50 within a category and about 0 across categories. Popularity is $\pi_i \propto \exp(s_{\text{pop}}\zeta_i)$ with $\zeta_i \sim \mathcal{N}(0,1)$, $s_{\text{pop}} = 1$, normalised within each category. The catalogue is regenerated for every seed.

3.2 Latent interests and drift

$z_t \in \Delta^{C-1}$. Base interest $z_A \sim \text{Dirichlet}(\kappa \mathbf{1}_C)$ with $\kappa \in \{0.1 \text{ (focused)}, 1.0 \text{ (spread)}\}$. The expected largest component is 0.67 and 0.29 respectively, and about 2.0 and 5.2 categories hold 80% of the mass. The shifted interest $z_B = Pz_A$ uses a half-swap permutation: the top $C/2$ categories map onto the bottom $C/2$ and vice versa. The mean $\text{TV}(z_A, z_B)$ is 0.99 (focused) and 0.64 (spread), and concentration is unchanged by the shift.

Scenario	Trajectory ($T = 200$)	Used by
Stable	$z_t = z_A$	H2, H4b
Abrupt	z_A for $t < 100$, then z_B	H1, H4a, H6
Gradual	$(1 - w_t)z_A + w_t z_B$, $w_t = \text{clip}((t - 80)/40, 0, 1)$	H1, H4a, H6
Recurrent	z_A, then z_B from $t = 70$, back to z_A from $t = 140$	H1, H4a, H6
Noisy	$z_t = z_A$, click noise $\eta = 0.3$	H3

Table 1: Drift scenarios. Shift times are fixed so that post-shift windows line up across users.

3.3 Interaction model

At each step: a noise flag $n_t \sim \text{Bernoulli}(\eta)$ with $\eta = 0.05$ (0.3 in the noisy scenario); category $c_t \sim \text{Cat}(z_t)$ if $n_t = 0$, otherwise uniform; then item $P(i \mid c, u) \propto \pi_i^{\gamma} \exp(\lambda\, \hat{s}_i(u))$ within the category, where u is a hidden taste vector, $\hat{s}_i(u)$ the standardised $\langle \phi_i, u \rangle$, $\gamma = 1$, $\lambda = 0.5$. Repeats are allowed. The exact marginal is

$$P_t(i) = \left[(1-\eta) z_{t,c_i} + \eta/C\right] P(i \mid c_i, u). \tag{1}$$

The hidden taste ($\lambda > 0$) is information the recommender cannot model, so even the oracle is not right by construction. Observed categories follow $(1-\eta)z_t + \eta/C$, which creates a noise floor $\text{TV} = \eta\ \text{TV}(z_t, \text{uniform})$ for any tracker.

3.4 Leakage firewall and seeds

The simulator returns an `ObservedSequence` (item ids only) and a separate `GroundTruth`. Estimators and recommenders see only observed items, public item categories and embeddings, and popularity estimated from a separate training population of 1,000 users × 50 interactions per seed. A test checks that the observed type carries no ground-truth field. Seeds come from `SeedSequence` streams keyed by (seed, path). Scenarios reuse each user's random-number bank (common random numbers), so scenario comparisons are paired, and before the first switch the stable, abrupt and recurrent sequences are identical.

Seed sets. Tuning 0–4; evaluation 100–119 (used once, at Milestone 8); exploratory 200–219 (Milestone 9). Scale per seed: 2 κ × 5 scenarios × 100 users × 200 steps = 200,000 interactions.

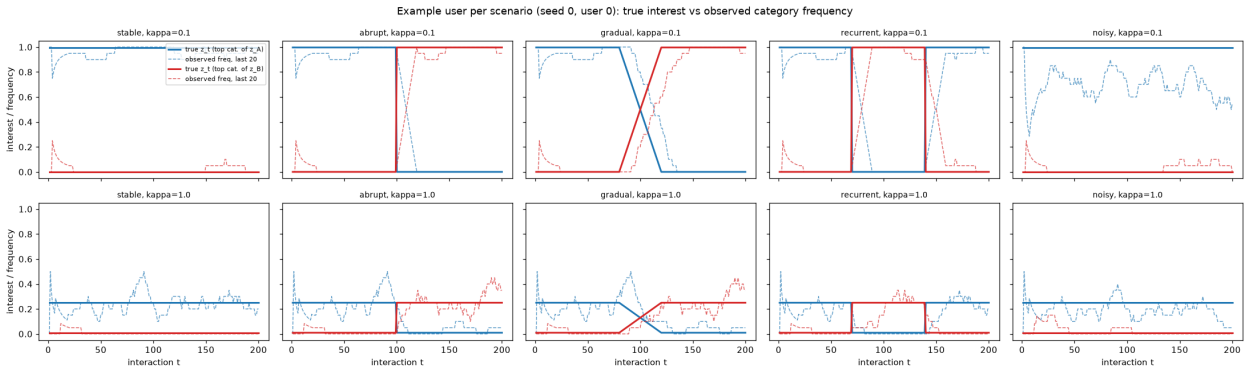


Figure 1: One example user per scenario and concentration: true z_t for the top categories of z_A and z_B against the rolling 20-interaction observed frequency. **TUNING SEEDS ONLY** *Source:* `results/m3_sim_check/20261026-192349/`

4 Methods

IMPLEMENTED + TESTED Code: `adaptdiv/estimators.py`, `relevance.py`, `similarity.py`, `dpp.py`, `recommenders.py`.

4.1 Interest estimators

With $x_t = e_{c_{i_t}}$, the one-hot category of the observed item:

$$\text{Cumulative mean (static):} \quad \hat{z}_t = \tfrac{1}{t}\textstyle\sum_{s\le t} x_s, \tag{2}$$

$$\text{Bias-corrected EWMA (adaptive):} \quad m_t = (1-\alpha)m_{t-1} + \alpha x_t,\ m_0 = 0, \quad \hat{z}_t = \frac{m_t}{1-(1-\alpha)^t}, \tag{3}$$

$$\text{Uniform-start EWMA (ablation):} \quad \hat{z}_t = (1-\alpha)\hat{z}_{t-1} + \alpha x_t,\ \hat{z}_0 = \mathbf{1}/C, \tag{4}$$

$$\text{Sliding window (ablation):} \quad \hat{z}_t = \tfrac{1}{\min(t,W)}\textstyle\sum_{s=\max(1,t-W+1)}^{t} x_s. \tag{5}$$

EWMA weights decay geometrically, with mean observation age $(1-\alpha)/\alpha$; a window has the same mean age when $W \approx 2/\alpha - 1$. The bias correction removes the uniform prior's influence on early estimates (D-27). Selected values: $\alpha = 0.05$ for the proposed estimator and $\alpha = 0.07$ for the uniform-start ablation, each tuned for tracking error on tuning seeds; $W = 40$.

4.2 Scores, similarity and selection

Base relevance is within-category popularity from the training population, $r_i = (\text{count}_i{+}1)/(\sum_{j\in c_i}\text{count}_j + n_c)$. The score is

$$s_i(t) = r_i\left(\hat{z}_{t,c_i} + \epsilon\right), \qquad \epsilon = 0.01, \tag{6}$$

and the similarity kernel is $S = \Phi\Phi^\top + 10^{-6}I$. All N items are candidates, and the list for step $t+1$ uses $\hat{z}_t$ only. The selection objective, our adaptation of Chen et al. (2018) (D-14), is

$$F(Y) = \theta \sum_{i\in Y} \log s_i + (1-\theta)\log\det S_Y, \qquad |Y| = K = 10. \tag{7}$$

For $\theta < 1$ this equals $(1-\theta)\log\det L_Y$ with $L = \text{Diag}(q)\,S\,\text{Diag}(q)$, $q_i = s_i^{\theta/(2(1-\theta))}$, so it is a valid DPP with log-linear quality. $\theta = 1$ is exactly Top-K, which puts both selectors on one scale. **Report wording:** Chen et al. use a linear form $\theta\sum r_i$. It failed at our score scale, so the log form is our adaptation and their results do not transfer directly.

Greedy MAP (Chen et al., Algorithm 1). Keep c_i and d_i^2; repeatedly pick $j = \arg\max_i\ \theta\log s_i + (1-\theta)\log d_i^2$, then update $e_i = (S_{ji} - \langle c_j, c_i\rangle)/d_j$, append e_i to c_i and set $d_i^2 \leftarrow d_i^2 - e_i^2$. The cost is $O(K^2N)$, measured at 0.12–0.18 ms per list. Greedy matched the exact optimum on 95%, 99.5% and 100% of 200 tiny instances (14 items, $k = 5$) at $\theta = 0.3, 0.6, 0.8$. This is tested on tiny instances only and does not bound the gap at $N = 500$. Call it *greedy approximate MAP*. DPP lists are sorted by score before evaluation, so both selectors use the same ranking rule.

Method	Profile	Selector
Popularity	none (global popularity)	Top-K
Static Top-K / Static DPP	cumulative mean	Top-K / DPP $\theta = 0.5$
Adaptive Top-K / Adaptive DPP	bias-corrected EWMA, $\alpha = 0.05$	Top-K / DPP $\theta = 0.5$
Oracle Top-K / Oracle DPP (diagnostic)	true z_{t+1}	Top-K / DPP $\theta = 0.5$
Ablations	uniform-start EWMA ($\alpha = 0.07$); window ($W = 40$)	both
Calibrated (M10, exploratory)	static or adaptive	Steck-style re-ranker,

Table 2: Methods (D-16). Oracles are ceilings, excluded from H6a.

5 Evaluation protocol

IMPLEMENTED + TESTED Code: `adaptdiv/metrics.py`, `evaluation.py`, `stats.py`, `confirmatory.py`. Decisions D-17, D-21, D-29, D-30.

5.1 Metrics

R_t is the list built from $\hat{z}_t$, and T_t is the set of distinct items among the next $h = 5$ clicks. Lists are scored for $t = 20, \ldots, T - h$.

- **NDCG@10** (primary relevance): binary gain, $\sum_{p=1}^{K} \mathbf{1}[R_t(p) \in T_t] / \log_2(p+1)$, divided by the ideal for $\min(|T_t|, K)$ hits.
- Recall@10; **expected utility** EU@10 $= \sum_{i \in R_t} P_{t+1}(i)$, computed by the simulator and used for the θ rule; new-item NDCG and EU as a sensitivity check on repeats.
- **Category coverage** (primary diversity): distinct categories in R_t divided by $\min(K, C)$. ILD, the mean pairwise $1 - \cos$, is descriptive only, because the DPP optimises it directly. Seen share is the fraction of R_t already clicked.
- **Estimation:** tracking error $\mathrm{TV}(\hat{z}_t, z_t)$; crossover lag, the first L such that $\hat{z}$ is closer to the new interest than to the old one for 5 consecutive steps, censored at the horizon; false-switch rate, the share of steps where $\arg\max \hat{z}$ changes while $\arg\max z$ does not.

The post-shift window is the 50 steps from each switch (abrupt 100, gradual midpoint 100, recurrent 70 and 140). Values are aggregated step → user → cell → seed. "Drift pooled" means the equal-weight mean over abrupt, gradual and recurrent and both κ.

5.2 Statistics and freeze

The unit of analysis is the seed-level mean, paired across methods (20 seeds). Tests are two-sided Wilcoxon signed-rank (smallest possible $p = 1.91 \times 10^{-6}$), with a percentile paired bootstrap 95% CI and Holm correction over 8 tests. SESOI: 3% relative for NDCG and EU, 2 interactions for lag, 0.02 for TV. "Practically meaningful" means the whole CI lies beyond the SESOI. Hypotheses, metrics, SESOI, α and θ were **frozen on 1 November 2026**, before any evaluation seed ran (`hypothesis.md`).

Test	Quantity (data)
H1a	crossover lag, cumulative − EWMA (drift pooled)
H1b	post-shift TV, cumulative − EWMA (drift pooled)
H2	TV, EWMA − cumulative (stable, $t > 20$)
H3	false-switch rate, EWMA − cumulative (noisy)
H4a	post-shift NDCG, Adaptive Top-K − Static Top-K (drift pooled)
H5a	coverage, Adaptive DPP − Adaptive Top-K (all scenarios)
H5b	relative NDCG cost of DPP (adaptive), focused − spread
H6b	(AdaDPP − StatDPP) − (AdaTop − StatTop) on post-shift NDCG, two-sided
H4b (outside Holm)	equivalence: whole CI of stable relative NDCG difference within ±3%
H6a (outside Holm)	Adaptive DPP not dominated on (post-shift NDCG, coverage) by Popularity, Static Top-K, Adaptive Top-K or Static DPP

Table 3: Confirmatory tests (D-30).

6 Tuning on tuning seeds

TUNING SEEDS ONLY

α for the estimator (pooled mean TV over all scenarios and both κ; `results/m4_ewma_init/20261101-140141/`):

θ for the DPP (D-29: the smallest θ with EU@10 loss $\leq 3\%$ against same-profile Top-K, for both profiles; `results/m7_theta/20261101-152049/`):

α	0.02	0.03	0.05	0.07	0.1	0.15	0.2	0.3	0.5
Uniform start	.2818	.2474	.2230	**.2216**	.2330	.2612	.2906	.3452	.4384
Bias-corrected	.2339	.2210	**.2138**	.2187	.2339	.2640	.2937	.3480	.4402

Table 4: Pooled tracking error. Cumulative mean: 0.2808. Best window $W = 40$: 0.2164.

θ	0.9	0.8	0.7	0.6	**0.5**	0.3
Static EU loss	−0.2%	−0.3%	−0.3%	0.2%	1.0%	6.9%
Adaptive EU loss	0.0%	0.1%	0.5%	1.3%	2.8%	10.3%
Adaptive coverage	0.332	0.345	0.360	0.380	0.406	0.490

Table 5: $\theta = 0.5$ selected; it sits close to the 3% limit.

7 Confirmatory results (Milestone 8)

CONFIRMATORY, EVALUATION SEEDS Run `results/m8_main/20261102-112410/`: seeds 100–119, every step, 812 s with 5 processes, git HEAD `f0fc858`, 29,440 records, no NaN. This was the first and only run on these seeds. Seed 100 rerun alone gave all 1,472 values identical.

Test	Mean difference	95% CI	Holm p	Verdict
H1a	+67.3 steps	[67.0, 67.6]	1.5×10^{-5}	Supported, beyond SESOI
H1b	+0.248 TV	[0.247, 0.249]	1.5×10^{-5}	Supported, beyond SESOI
H2	+0.045 TV	[0.045, 0.046]	1.5×10^{-5}	Supported, beyond SESOI
H3	+0.057	[0.055, 0.058]	1.5×10^{-5}	Supported
H4a	+0.101 NDCG (+131%)	[122%, 140%]	1.5×10^{-5}	Supported, beyond SESOI
H5a	+0.089 coverage	[0.086, 0.092]	1.5×10^{-5}	Supported
H5b	+0.0038	[0.0012, 0.0064]	0.024	Supported, very small
H6b	−0.0106 NDCG	[−0.0111, −0.0100]	1.5×10^{-5}	Significant: substitution
H4b	$\approx -1.8\%$	[−1.9%, −1.6%]	–	Equivalent within ±3%
H6a	–	–	–	Not dominated

Table 6: Confirmatory tests. 1.5×10^{-5} is the floor after Holm ($8 \times 1.91 \times 10^{-6}$).

Reading the numbers.

- **Adaptation gives the large gain.** Post-shift NDCG rises from 0.0787 to 0.1794. After a shift, Static Top-K falls below even Popularity (0.0787 vs 0.1026).
- **The DPP's effect is small and changes sign**: +0.0074 with the static profile and −0.0032 with the adaptive one. That is the H6b substitution.
- Adaptive DPP vs Adaptive Top-K: NDCG cost 1.8%, coverage +0.089 (0.299 to 0.388). The overall EU loss is 3.04%, just above the 3% limit met on tuning seeds (2.8%); it is reported as the out-of-sample value.
- **Estimation matters more than selection.** Oracle Top-K beats Adaptive Top-K by 0.055 post-shift NDCG, about 17 times the DPP's cost.
- H2 and H3 are predicted costs of adaptation, and both appeared.

Ablations (descriptive, outside Holm; variant − proposed, drift pooled, post-shift NDCG). Uniform-start EWMA ($\alpha = 0.07$): +0.0114 [0.0111, 0.0118] Top-K, +0.0113 DPP, coverage −0.022 and −0.030. Window(40): −0.0179 [−0.0184, −0.0175] Top-K, −0.0154 DPP.

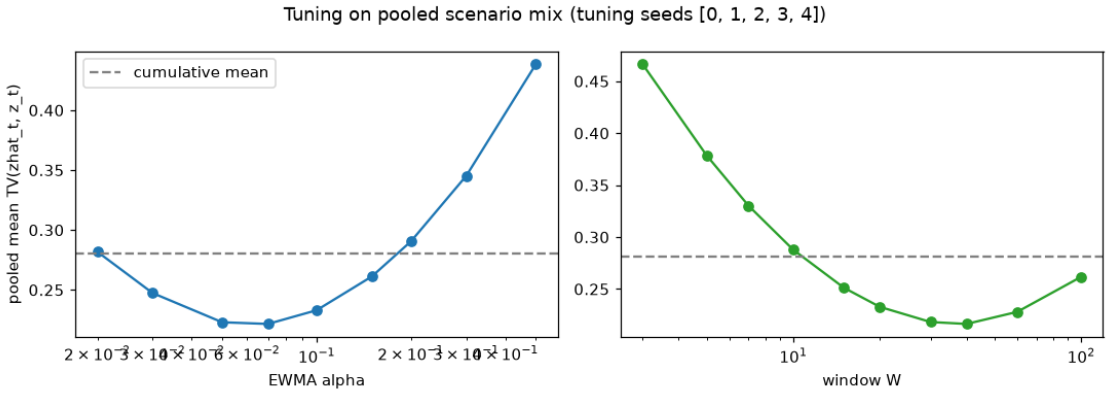


Figure 2: Estimator tuning curves (uniform-start EWMA and windows; the bias-corrected curve is in the table above). **TUNING SEEDS ONLY** *Source:* `results/m4_estimators/20261028-121944/`

Method	Post-shift (drift pooled)			Overall (all scenarios)		
	NDCG	EU	Coverage	NDCG	Coverage	Seen share
Popularity	0.1026	0.1281	0.745	0.1029	0.745	0.490
Static Top-K	0.0787	0.1027	0.358	0.1774	0.328	0.805
Adaptive Top-K	0.1794	0.2129	0.341	0.2086	0.299	0.756
Static DPP	0.0861	0.1175	0.463	0.1762	0.421	0.787
Adaptive DPP	0.1762	0.2064	0.444	0.2049	0.388	0.744
Oracle Top-K	0.2348	0.2736	0.294	0.2281	0.287	0.720
Oracle DPP	0.2304	0.2648	0.376	0.2241	0.366	0.710

Table 7: Pooled means over 20 seeds. Seed SD of Adaptive DPP post-shift NDCG: 0.012.

8 Analysis and exploratory follow-ups (Milestone 9)

8.1 Per-scenario breakdown of the confirmatory run

Descriptive, outside Holm, from the saved records. **Pooling hid a per-cell cost.** H4b meets the ±3% rule pooled, but Adaptive Top-K's overall NDCG is lower than Static Top-K's by:

The frozen H4b verdict stands, and these cells should be reported beside it.

8.2 α versus EWMA start

EXPLORATORY Seeds 200–219, 553 s, `results/m9_alpha_start/20261102-151847/`. A 2×2 of $\alpha \in \{0.05, 0.07\}$ and start $\in$ {bias-corrected, uniform}:

The ablation's gain came from α, not from the start: the start's leftover weight $(1-\alpha)^t$ is under 3% by the first shift. A larger α helps after shifts and costs when interests are stable, so neither value dominates. The frozen $\alpha = 0.05$ stays.

8.3 Sensitivity to θ and click noise θ

EXPLORATORY Seeds 200–219, 1,614 s, `results/m9_sensitivity/20261102-164513/`. Here η also changes the training population, and the noisy scenario runs only at the default η. These sweep runs reproduce the α study's values exactly where they overlap (5,120 of 5,120). CIs are not corrected for multiplicity.

Findings.

- **Substitution is robust**: negative, with the CI excluding zero, at every θ and η tested and in both κ groups. It grows with DPP strength and shrinks a little with noise.

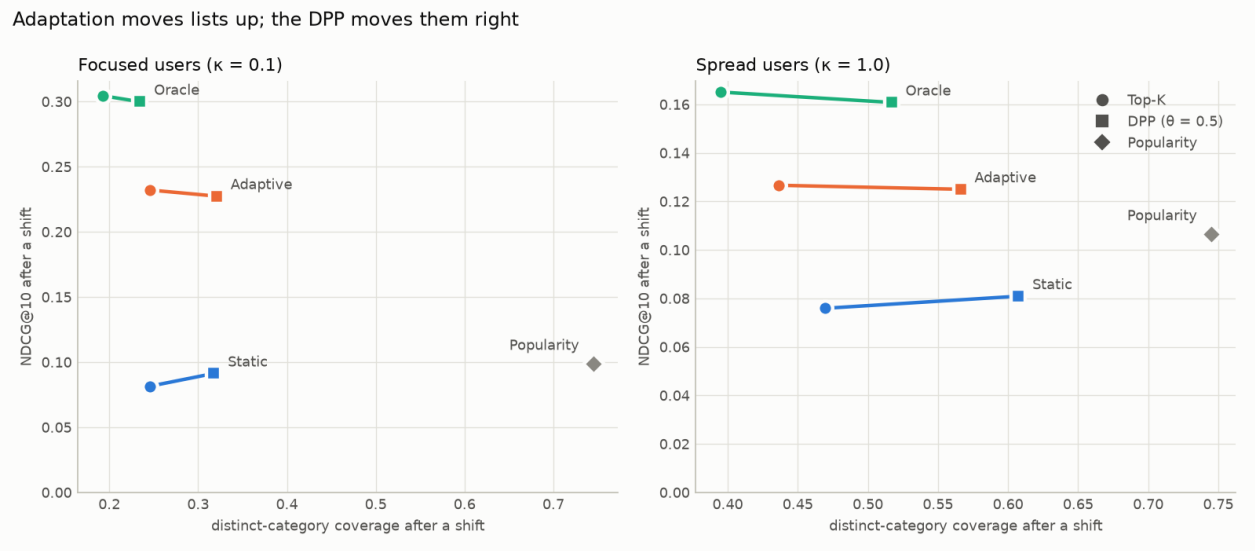


Figure 3: Post-shift NDCG against coverage. Arrows go from each Top-K (circle) to its DPP (square). **CONFIRMATORY, EVALUATION SEEDS** *Source:* `results/m9_analysis/20261102-151914/tradeoff.png`

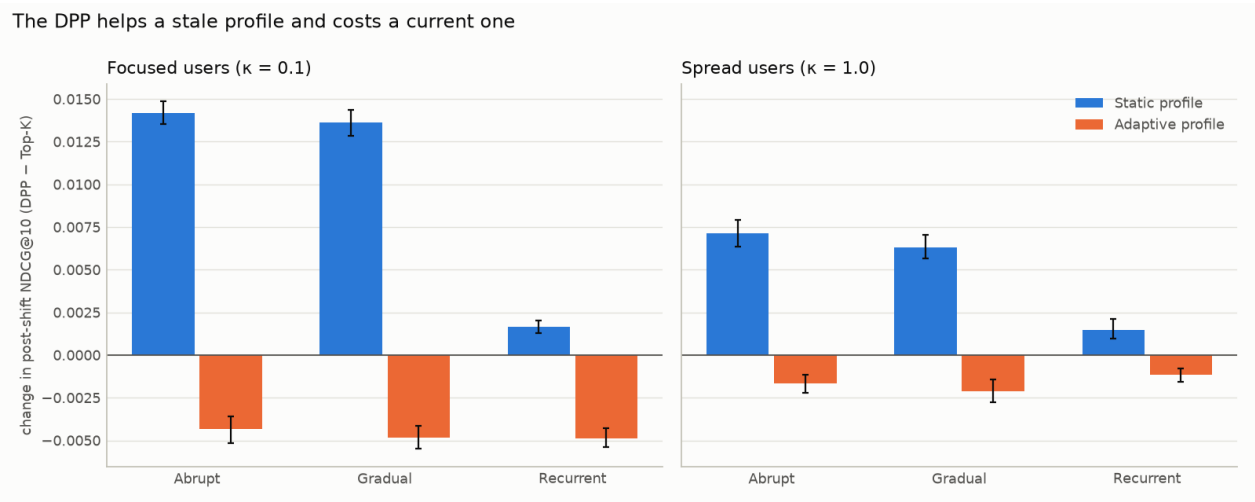


Figure 4: DPP − Top-K on post-shift NDCG per drift scenario, with paired 95% CIs. Positive for the static profile in all 6 cells (+0.0015 to +0.0142), negative for the adaptive profile in all 6 (−0.0011 to −0.0049). **CONFIRMATORY, EVALUATION SEEDS** (descriptive breakdown)

Method	Focused ($\kappa = 0.1$)			Spread ($\kappa = 1.0$)		
	Post NDCG	Post cov.	Stable NDCG	Post NDCG	Post cov.	Stable NDCG
Popularity	0.0987	0.745	0.0985	0.1065	0.745	0.1058
Static Top-K	0.0815	0.245	0.3117	0.0759	0.470	0.1610
Adaptive Top-K	0.2321	0.246	0.3093	0.1266	0.436	0.1550
Static DPP	0.0913	0.318	0.3062	0.0809	0.607	0.1571
Adaptive DPP	0.2274	0.321	0.3040	0.1250	0.566	0.1523
Oracle Top-K	0.3045	0.192	0.3131	0.1651	0.395	0.1668
Oracle DPP	0.2999	0.235	0.3081	0.1608	0.517	0.1627

Table 8: Means by concentration (computed from `records.csv`). The interaction is -0.0145 for focused and -0.0066 for spread users.

κ	Scenario	Cumulative mean		EWMA ($\alpha = 0.05$)	
		TV (post)	lag / FS	TV (post)	lag / FS
0.1	stable	0.075	FS 0.008	0.103	FS 0.031
0.1	abrupt	0.790	98.7	0.386	13.2
0.1	gradual	0.685	98.2	0.311	16.5
0.1	recurrent	0.587	53.9	0.379	12.8
0.1	noisy	0.227	FS 0.008	0.243	FS 0.034
1.0	stable	0.116	FS 0.023	0.178	FS 0.100
1.0	abrupt	0.515	93.5	0.297	13.9
1.0	gradual	0.448	93.7	0.253	16.8
1.0	recurrent	0.385	52.2	0.295	13.5
1.0	noisy	0.161	FS 0.026	0.211	FS 0.114

Table 9: Estimation per cell. Stable and noisy rows: overall TV and false-switch rate (FS). Drift rows: post-shift TV and crossover lag in steps. The cumulative mean's lag is censored at the horizon for 64% of abrupt and gradual cases and 29% of recurrent ones, so H1a understates the true gap.

- **The DPP cannot replace adaptation.** Its largest static gain is 0.0118, against 0.1003 from adaptation; Static DPP at $\theta = 0.3$ (0.0935) stays far below Adaptive Top-K (0.1819).
- **Diversity gets expensive quickly**: for the adaptive profile, $\theta = 0.7$ costs 0.33% of post-shift NDCG for $+0.046$ coverage, while $\theta = 0.3$ costs 6.6% for $+0.185$. For spread users at $\theta = 0.7$ the cost is -0.0001 $[-0.0003, +0.0001]$.
- **Adaptation survives noise.** Its gain falls about 40% from $\eta = 0$ to 0.3 and stays large. Spread users exceed a 3% stable cost at every η, even $\eta = 0$, so that cost is not caused by noise alone.
- Static post-shift NDCG rises slightly with η (0.0803 to 0.0880). A plausible explanation, untested: noise flattens the target clicks, so a stale list misses slightly less.

$\theta = 0.7$ **looks attractive here, but it was seen after the fact.** D-29 fixed the rule in advance, so changing θ would need confirmation on fresh seeds.

9 Calibration baseline (Milestone 10, D-19)

EXPLORATORY Planned since Milestone 1 to answer "is a DPP needed at all?", built after the Steck paper was read in full. Code: `adaptdiv/calibration.py`.

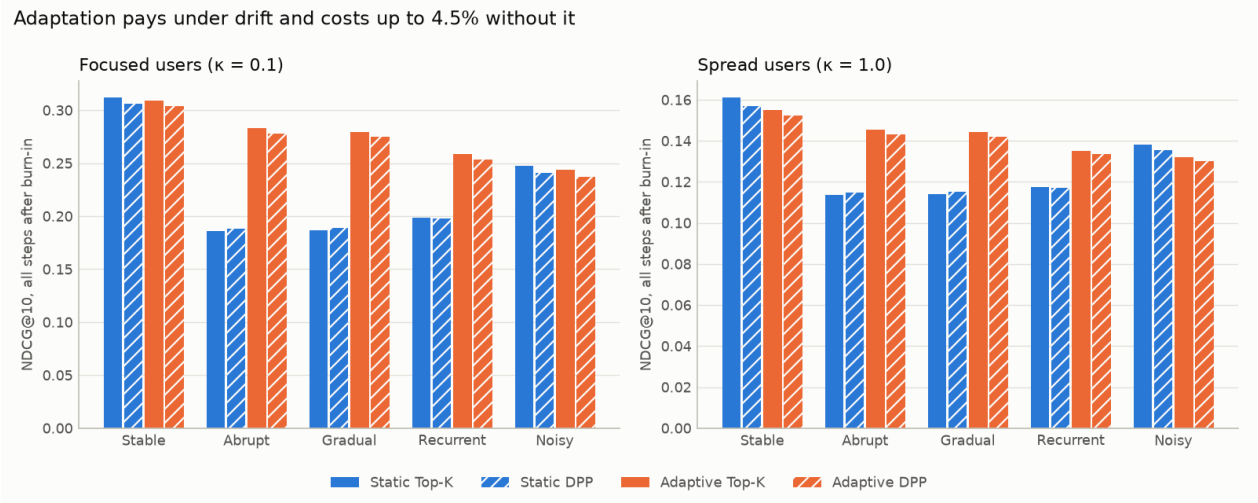


Figure 5: Overall NDCG of the four factorial methods per scenario; hatching marks the DPP. **CONFIRMATORY, EVALUATION SEEDS** (descriptive breakdown)

	Stable	Noisy
Focused	−0.8%	−1.4%
Spread	−3.7% (−0.0060, CI [−0.0064, −0.0054])	−4.5% (−0.0062, CI [−0.0067, −0.0058])

Method (after Steck (2018), Eq. 6, with our adaptations). For the method's own profile $p = \hat{z}_t$ and list R with category mix q_R:

$$R_t = \arg\max_{|R|=K} (1-\lambda) \sum_{i \in R} \log s_i - \lambda \, \mathrm{KL}\big(p \,\|\, \tilde{q}_R\big), \qquad \tilde{q}_R = (1-a)\, q_R + a\, p, \;\; a = 0.01, \tag{8}$$

built greedily one item at a time. $\lambda = 0$ is Top-K. Adaptations: $\log s_i$ instead of s_i (the paper allows a monotone transform), no rank weights and no diversity prior (as in the paper's experiments). Steck's own target allows recency weights (his Eq. 2), so an EWMA target is within his framework. New descriptive metric for every method: miscalibration $\mathrm{KL}(z_{t+1} \,\|\, \tilde{q}_R)$ against the *true* interest.

Choosing λ **TUNING SEEDS ONLY** The same rule as θ: the most calibrating λ with EU loss $\leq 3\%$ for both profiles (`results/m10_calibration_lambda/20261103-160304/`).

Comparison **EXPLORATORY** Seeds 200–219, 324 s, `results/m10_calibration/20261103-160313/`. Overall EU is equal for the two adaptive diversifiers (0.2405 and 0.2405), so this compares them at a matched relevance budget. The DPP and Top-K values reproduce the sensitivity run exactly (10,240 of 10,240).
Interaction (adaptation × selector) on post-shift NDCG: DPP −0.0105 [−0.0111, −0.0100]; calibrated −0.0133 [−0.0137, −0.0128].

Findings.

- **At the same relevance budget, the DPP added nothing that calibration did not.** With the adaptive profile NDCG and EU tie, and calibration gives about 0.08–0.10 more coverage, lower miscalibration and even slightly higher ILD, the measure the DPP optimises.
- By H6a's rule Adaptive Calibrated does not formally dominate Adaptive DPP, only because NDCG is tied.
- **Substitution is not DPP-specific.** Calibration shows it too, slightly more strongly.

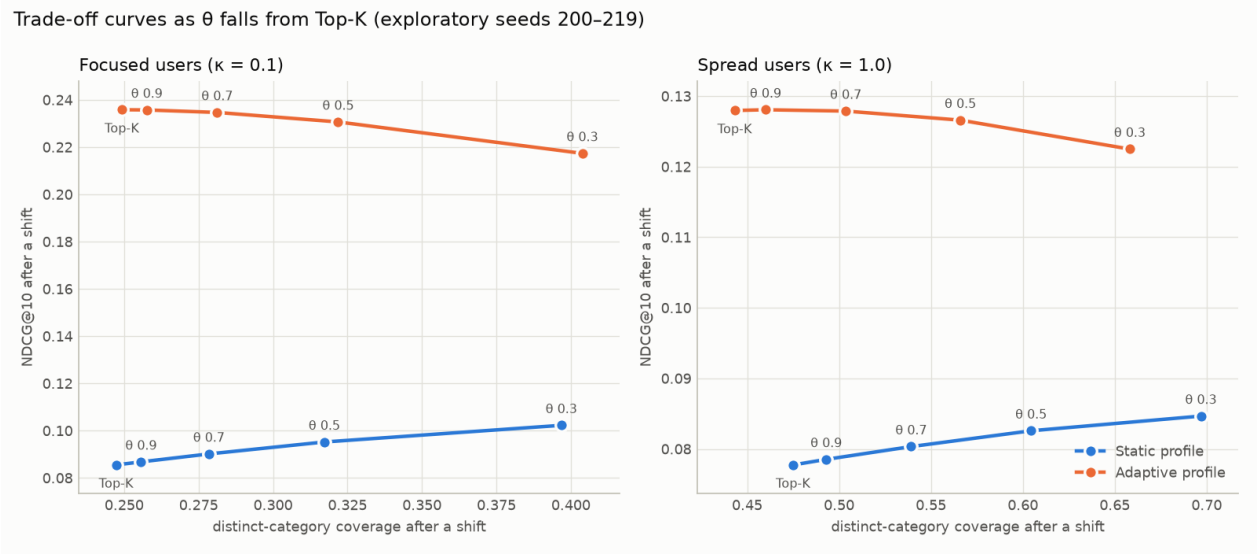


Figure 6: Post-shift NDCG against coverage as θ falls from Top-K, per profile and concentration. **EXPLORATORY** *Source:* `results/m9_sensitivity/20261102-164513/frontier.png`

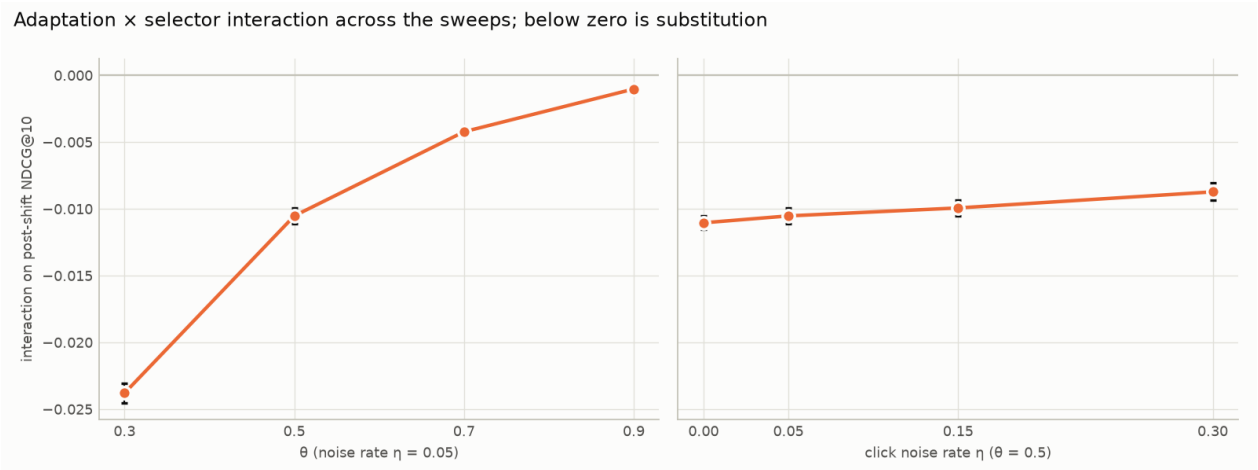


Figure 7: Adaptation × selector interaction with 95% CIs along the θ and η sweeps; below zero is substitution. **EXPLORATORY**

Top-K	α 0.07 − 0.05	Uniform − bias-corrected start
Post-shift NDCG	+0.0114 [0.0112, 0.0116]	+0.00002 [0.00001, 0.00002]
Post-shift coverage	−0.0225 [−0.0235, −0.0215]	+0.0003
Stable NDCG	−0.0024 [−0.0025, −0.0023]	+0.00002
All-scenario NDCG	+0.0008 [0.0007, 0.0009]	+0.00002

θ	DPP effect, static	DPP effect, adaptive	Interaction (95% CI)	Adaptive cov. gain	Adaptive EU loss
0.9	+0.0010	+0.0000	−0.0010 [−0.0011, −0.0009]	+0.013	0.03%
0.7	+0.0036	−0.0006	−0.0042 [−0.0045, −0.0039]	+0.046	0.62%
0.5	+0.0073	−0.0033	−0.0105 [−0.0111, −0.0100]	+0.098	3.12%
0.3	+0.0118	−0.0120	−0.0238 [−0.0245, −0.0231]	+0.185	10.84%

Table 10: θ sweep at $\eta = 0.05$: post-shift NDCG, drift pooled, both κ.

- A plausible reason, untested: embeddings are clustered by category (cross-category cosine about 0), so embedding diversity is mostly category diversity, which calibration targets directly. A catalogue with real within-category variety might favour a DPP; this simulator cannot show it.

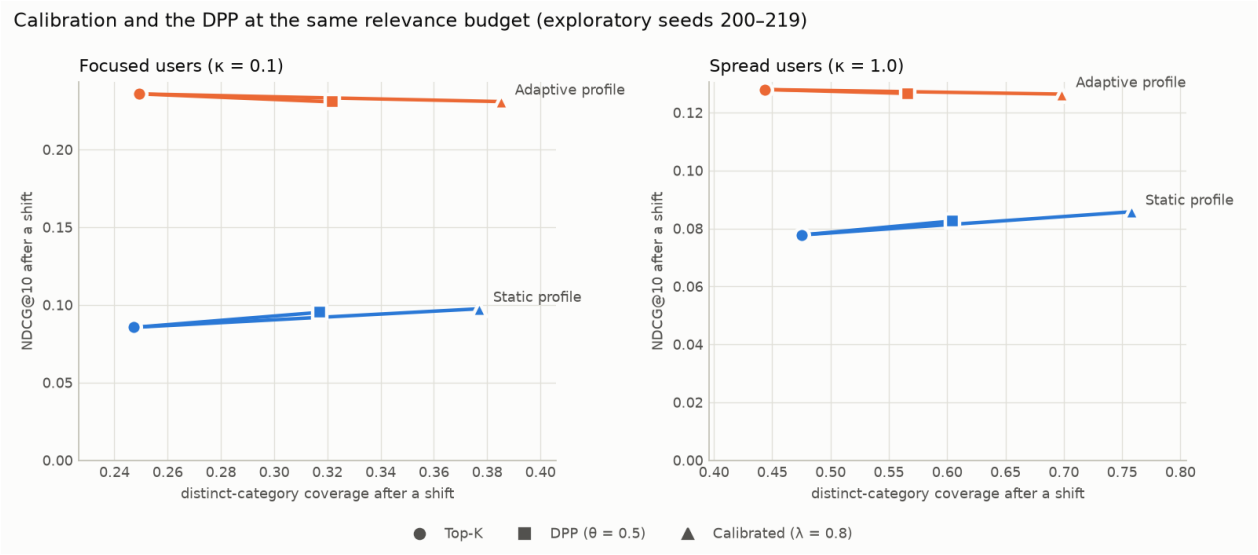


Figure 8: Post-shift NDCG against coverage for Top-K (circle), DPP $\theta = 0.5$ (square) and calibrated $\lambda = 0.8$ (triangle). **EXPLORATORY** *Source:* `results/m10_calibration/20261103-160313/calibration_tradeoff.png`

10 Negative results

- Diversification did not help the adaptive profile's relevance: the interaction is substitution, not complementarity, at every θ and η tested.
- The proposed configuration is not the best adaptive setting for post-shift NDCG: $\alpha = 0.07$ is about 6% higher, though it costs stable NDCG and coverage.
- Adaptive DPP's out-of-sample EU loss (3.04%) exceeded the 3% limit met on tuning seeds.
- Adaptation has real costs: worse stable tracking (H2), more false switches (H3), and lower stable NDCG, up to 3.7–4.7% for spread users.
- After a shift, the static profile is worse than plain Popularity.

η	Adaptation gain (post NDCG)	Stable cost, all	Focused	Spread	Interaction (95% CI)
0	+0.1079	−1.59%	−0.64%	−3.43%	−0.0110 [−0.0115, −0.0105]
0.05	+0.1003	−1.68%	−0.69%	−3.58%	−0.0105 [−0.0111, −0.0100]
0.15	+0.0861	−1.92%	−0.83%	−3.95%	−0.0099 [−0.0105, −0.0093]
0.3	+0.0648	−2.51%	−1.26%	−4.73%	−0.0087 [−0.0094, −0.0081]

Table 11: η sweep: adaptation columns use Top-K, the interaction uses $\theta = 0.5$.

λ	0.5	**0.8**	0.9	0.95	0.98	0.99	0.995
Static EU loss	−0.5%	0.6%	2.6%	5.0%	7.3%	8.3%	8.9%
Adaptive EU loss	0.4%	**2.8%**	5.6%	8.3%	10.5%	11.4%	12.0%

Table 12: $\lambda = 0.8$ selected; like θ, it sits close to the limit.

- $\theta = 0.3$ costs the adaptive profile about 6.6% post-shift NDCG and 10.8% EU.
- At a matched relevance budget the DPP was no better than a calibrated re-ranker, which gave more coverage, lower miscalibration and slightly higher ILD (§9).

11 Limitations and threats to validity

1. **Synthetic only.** The generator is category-driven and the scorer is category-weighted, which favours category-level profiles. Mitigations: a hidden taste $\lambda > 0$, a stable scenario where adaptation should not help, two κ levels, and a noise sweep.
2. **Open loop.** Recommendations never change behaviour, so there is no feedback loop and no exposure effect.
3. **Repeats allowed.** 69–87% of Top-K list items were already clicked (tuning check), and repeats are 33–64% of targets. The results apply to domains with repeat consumption, and new-item metrics are a sensitivity check.
4. **Fixed shift times** and a half-swap shift; no randomised τ. A pooled tuning mix with 2 of 5 scenarios without drift shapes α.
5. **Adapted DPP objective** (log form), so Chen et al.'s results do not transfer; greedy quality is verified on tiny instances only.
6. θ **near the rule limit** (2.8% on tuning, 3.04% out of sample).
7. **Censored lags** understate H1a.
8. **Noise floor**: estimators track $(1 - \eta)z + \eta/C$, not z.
9. **Narrow CIs** reflect variation between 20 paired seeds with shared randomness, not generalisation. Exploratory CIs are uncorrected.
10. **Exploratory follow-ups** were chosen after seeing confirmatory results, and seeds 200–219 are shared across the follow-up studies.
11. **Calibration baseline is exploratory**: added after the confirmatory run, on seeds 200–219; λ's grid has no value between 0.8 and 0.9.
12. **Literature** mostly verified at metadata or abstract level; targeted search.

12 Pending items

- Done on 3 November 2026: the calibration baseline (D-19), §9.

Calibrated − DPP	Post NDCG	Post coverage	Overall coverage	Miscalibration	ILD (abs.)
Static	+0.0027 [0.0023, 0.0032]	+0.107	+0.091	−0.283	0.857 vs 0.845
Adaptive	−0.0000 [−0.0003, 0.0003]	+0.098	+0.083	−0.222	0.840 vs 0.828

Table 13: Pooled over κ; the same direction holds in both κ groups.

- **PENDING** Deadline (D-07) not recorded.
- Not done by design: real-data validation (D-31); a randomised shift-time sensitivity check; λ and $\kappa = 0.3$ sensitivity.

13 Reproducibility

Repository https://github.com/Shambo-Roy/ire-adaptive-dpp-rec. Python 3.12.10, NumPy 2.5.2, SciPy 1.18.0, matplotlib 3.11.1; CPU only (Windows 11, 20 logical CPUs). The test suite has 188 tests (`python -m pytest -q`).

Milestone	Command	Output
M3	`python experiments/sim_check.py`	`results/m3_sim_check/`
M4	`python experiments/estimator_study.py`; `ewma_init_comparison.py`	`results/m4_*`
M5	`python experiments/relevance_check.py`	`results/m5_relevance/`
M6	`python experiments/dpp_check.py`	`results/m6_dpp/`
M7	`python experiments/select_theta.py`	`results/m7_theta/`
M8	`python experiments/main_experiment.py` (about 14 min)	`results/m8_main/`
M9	`python experiments/m9_figures.py`	`results/m9_analysis/`
M9	`python experiments/alpha_start_study.py` (about 9 min)	`results/m9_alpha_start/`
M9	`python experiments/sensitivity_study.py` (about 27 min); `sensitivity_figures.py`	`results/m9_sensitivity/`
M10	`python experiments/select_calibration.py` (about 2 min)	`results/m10_calibration_lambda/`
M10	`python experiments/calibration_study.py -lam 0.8` (about 5.5 min); `calibration_figures.py`	`results/m10_calibration/`

Every run folder stores `config.json`, the seed list, NumPy version, git HEAD and status, and its runtime. The main experiment refuses to run if `adaptdiv/` differs from HEAD.

14 Claim-to-evidence map

Claim	Evidence
EWMA recovers faster and tracks better after drift	H1a, H1b; estimation table (§7)
Adaptation costs accuracy when stable or noisy	H2, H3, H4b; per-scenario table (§8)
Adaptation raises post-shift relevance a lot	H4a; Table 7; Fig. 3
The DPP raises coverage at a small relevance cost	H5a; Table 7; Fig. 6
The DPP's cost is slightly larger for focused users	H5b (very small)
Adaptation and DPP substitute	H6b; Fig. 4; Fig. 7
Adaptive DPP is not dominated	H6a
The DPP adds nothing over calibration here	§9; Fig. 8
The $\alpha = 0.07$ ablation gain comes from α	§8, $\alpha \times$ start
Estimation matters more than selection	Oracle gap 0.055 vs DPP cost 0.0032

A Viva preparation

Design

Why a simulation and not a real dataset?
Real logs do not reveal the true interest z_t or when it changed. Without them we cannot measure tracking error, lag or false switches, and cannot separate estimation from selection. The price is external validity.

Doesn't the simulator favour your method?
Partly: generator and scorer are both category-based. Mitigations: hidden taste the recommender cannot model, a stable scenario where adaptation should lose (it did, H2 and H4b), a noisy scenario, and reporting every scenario. The oracle still misses, because of $\lambda > 0$.

Why open-loop?
To isolate estimation and selection effects. Closed-loop feedback would mix them with exposure effects. It is a stated limitation.

Why a half-swap shift?
It keeps concentration constant, so drift is not confounded with a change in how focused the user is. A top-2 swap was too weak for spread users (mean TV 0.36).

Why common random numbers?
Scenario differences then come only from the trajectory, and comparisons are paired, which lowers variance.

Why not a neural session model?
The question is about the profile and the selector. A transparent plug-in scorer keeps the effects interpretable, and simple models are often competitive (Ludewig and Jannach, 2018).

Estimation

Why bias-corrected EWMA?
With a uniform start, the stable-scenario gap mixed two causes: forgetting and the prior. Bias correction removes the prior. It also shifted tuning from $\alpha = 0.07$ to 0.05.

What is the EWMA's mean memory?
Mean observation age $(1 - \alpha)/\alpha = 19$ steps at $\alpha = 0.05$. A window with the same mean age has $W \approx 2/\alpha - 1 = 39$.

Why does EWMA beat a window at matched memory?
EWMA weights the newest observations most, so it crosses over sooner: 9.3 steps for EWMA(0.07) against 13.5 for a window of the same mean age (28), on tuning seeds.

Why is H1a's 67 steps an underestimate?
The cumulative mean never crosses over within the horizon for 64% of abrupt and gradual cases; those lags are recorded at the horizon.

Should you switch to $\alpha = 0.07$?
Not on this evidence. It gains about 6% after shifts and loses when stable. The right value depends on how often interests change, and it would need confirming on fresh seeds.

DPP

What does $\det S_Y$ measure?
The squared volume spanned by the selected embeddings. Near-duplicates make it small, so the determinant rewards spread.

Why greedy and not exact MAP?
Exact MAP is NP-hard. Greedy with Chen et al.'s updates costs $O(K^2N)$, about 0.15 ms per list, and found the exact optimum in 95–100% of tiny tests.

Why your own objective form?
Chen's linear form ignored relevance at our score scale. The log form makes $\theta = 1$ exactly Top-K and is still a valid DPP.

Why sort DPP lists before NDCG?
The DPP picks a set. Greedy order is an artefact that cost up to 0.9% NDCG in a probe, and sorting gives both selectors the same ranking rule.

Why coverage and not ILD as the diversity metric?
The DPP optimises embedding similarity directly, so ILD gains are partly true by construction.

How was θ chosen?
Before evaluation, on tuning seeds: the most diverse θ whose EU loss was at most 3% for both profiles.

Results

What does substitution mean?
The DPP helps less when the profile is already current. It helps a stale profile by hedging, since diversity spreads slots onto categories the user may have moved to. It costs a current profile relevance.

So is the DPP useless?
No. It buys coverage for a small relevance cost, but it does not improve relevance for a current profile, and it cannot replace adaptation (0.0118 vs 0.1003).

Why does the static profile lose to Popularity after a shift?
It still recommends the old categories, which the user has left. Popularity at least spreads across categories.

Why does the static DPP's EU rise while the adaptive DPP's falls?
A list built on wrong weights has room to improve. A list built on nearly right weights can only give up relevance.

Why are the confidence intervals so narrow?
They cover 20 paired seeds with shared randomness and 100 users per cell. They describe seed variability in this simulator, not generalisation.

Why seed-level tests and not user-level?
Users within a seed share a catalogue and are not independent. The seed is the unit that is replicated.

Why does H4b pass while spread users lose 3.7–4.5%?
H4b was frozen as a pooled test. The per-cell breakdown is descriptive and reported beside it rather than replacing it.

Why use new seeds for follow-ups?
The follow-up questions came from looking at evaluation results. Answering them on the same seeds would mix discovery with confirmation.

Did you compare against a non-DPP diversifier?
Yes, a calibrated re-ranker after Steck, chosen by the same EU rule. At equal relevance it beat the DPP on coverage, miscalibration and even ILD. In this simulator the DPP is not needed for these goals; it might matter where items vary within categories.

Why $\lambda = 0.8$ and not Steck's 0.99?
Our score term is a sum of log scores over 10 items and 10 categories, so the scales differ. λ was chosen by the same rule as θ.

What would you do next?
Real logs with proxy drift labels; a closed-loop simulator; adaptive-rate estimators such as ADWIN (Bifet and Gavaldà, 2007); randomised shift times.

Key numbers to remember

- Adaptation: +0.101 post-shift NDCG (+131%); lag gap +67 steps; TV gap 0.248.
- Cost of adaptation: stable TV +0.045; false switches 0.074 vs 0.017 in the noisy scenario; stable NDCG about −1.8%.
- DPP: coverage +0.089; NDCG cost 1.8%; EU loss 3.04%.
- Interaction: −0.0106 (substitution). Oracle gap: 0.055.
- Calibration vs DPP (adaptive, equal EU): NDCG tie, coverage +0.083 overall.
- Scale: 20 evaluation seeds × 2 κ × 5 scenarios × 100 users × 200 steps.

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